

# Faithfulness Verification and Hallucination Mitigation in LLM Generation

David Miller, Maya Lin, Arthur Pendelton — Center for Reliable AI Research

## Abstract

Large Language Models often generate plausible-sounding but factually ungrounded statements, commonly known as hallucinations. In academic research synthesis, unverified claims can lead to severe misinformation. We introduce an automated claim-level Verification Agent that breaks candidate answers into atomic propositional claims and validates each claim against retrieved source chunks using Natural Language Inference (NLI) classification: SUPPORTED, PARTIALLY\_SUPPORTED, or UNSUPPORTED.

## 1. The Hallucination Problem in Academic RAG

Even when authoritative documents are injected into the LLM context window, generative models often hallucinate by extrapolating beyond the evidence, attributing findings to the wrong authors, or fusing contradictory claims. Strict verification is therefore non-negotiable for academic reliability.

## 2. Verification Pipeline Architecture

Our verification framework operates in three sequential phases:

1. Claim Decomposition: Extracts atomic factual statements from the draft response.
2. Evidence Alignment: Matches each claim to the exact source chunk containing relevant tokens.
3. Entailment Verification: Classifies whether the chunk logically entails the claim, assigning a confidence score [0.0 to 1.0].

### 3. Experimental Results on Faithfulness Benchmarks

Evaluating our verification module against 500 expert-annotated research claims yielded an F1-score of 91.4% in detecting factual hallucinations. Filtering out UNSUPPORTED claims before user presentation improved user trust ratings by 68% on a 5-point Likert scale.

### 4. Strengths and Practical Benefits

- **Strict Grounding:** Ensures no fabricated facts or synthetic numbers reach the final answer.
- **Transparent Citations:** Every accepted claim links directly to its parent document and page index.
- **Graceful Uncertainty Handling:** If evidence is absent, the system explicitly declares insufficient evidence rather than guessing.

### 5. Limitations

The verification agent relies on the quality of the retrieved context. If the retrieval module omits key context, valid true claims may be mistakenly classified as UNSUPPORTED due to context deficiency.

### 6. Conclusion

Post-generation verification is essential to achieve factual integrity in GenAI research synthesis systems.